Second order linear homogeneous equations

At this point, we want to turn our attention away from first order differential equations, and toward second order equations.

Second order equations can take many forms, but we want to start by looking at second order linear equations that have constant coefficients. Second order linear equations can be homogeneous or nonhomogeneous, and we'll start here with the homogeneous variety. The form we're looking for is

$$y'' + Ay' + By = 0$$

This is a second order equation because it includes a second derivative. It's a linear equation because we only see y and its derivatives, and not any other function of y, like $\cos y$, e^y, y^3, etc. The coefficients A and B are constants, and the equation is homogeneous because of the zero-value on the right side.

Finding the solution

To find the solution that satisfies a second order linear homogeneous equation with constant coefficients, we always start with the associated **characteristic equation**,

$$r^2 + Ar + B = 0$$

In other words, to change the differential equation into its matching characteristic equation, we substitute $y'' = r^2$, $y' = r$, and $y = 1$. Once we've

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done that, we realize that the characteristic equation is a simple quadratic equation, and we can solve it using methods we learned in algebra, including factoring, completing the square, or the quadratic formula.

Remember that quadratic equations will have one of three types of roots: distinct real roots, equal real roots, or complex conjugate roots. Distinct real roots are real-numbered roots that are unequal; equal real roots are real-numbered roots that are equivalent. Complex conjugate roots are complex-numbered roots.

The general solution to the differential equation is determined by the type of roots r_1 and r_2 that we find from the characteristic equation.

Roots	**General solution**
Distinct real roots	$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
Equal real roots	$y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$
Complex conjugate roots	$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$
	where $r = \alpha \pm \beta i$

The general solution will contain the constants c_1 and c_2, but we can find the values of c_1 and c_2 if we're given initial conditions. Let's try an example so that we can see how to find the general solution.

Example

Find the general solution to the second order equation.

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$$y'' + 5y' + 6y = 0$$

The characteristic equation associated with the differential equation is

$$r^2 + 5r + 6 = 0$$

This quadratic equation is easily factorable,

$$(r + 2)(r + 3) = 0$$

which means the roots will be

$$r_1 + 2 = 0 \qquad r_2 + 3 = 0$$

$$r_1 = -2 \qquad r_2 = -3$$

These are real-numbered unequal roots, so we call them distinct real roots. Therefore, the general solution to the second order equation is

$$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$y(x) = c_1 e^{-2x} + c_2 e^{-3x}$$

Realize here that it makes no difference whether we assign $r_1 = -2$ and $r_2 = -3$, or $r_1 = -3$ and $r_2 = -2$. In other words, both of these are acceptable general solutions:

$$y(x) = c_1 e^{-2x} + c_2 e^{-3x}$$

$$y(x) = c_1 e^{-3x} + c_2 e^{-2x}$$

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The constants c_1 and c_2 are simply placeholders until we can apply initial conditions, so we don't have to worry about how we name the roots.

Let's look at an example with equal real roots.

Example

Find the general solution to the second order equation.

$$y'' + 6y' + 9y = 0$$

The characteristic equation associated with the differential equation is

$$r^2 + 6r + 9 = 0$$

This quadratic equation is easily factorable,

$$(r + 3)(r + 3) = 0$$

which means the roots will be

$$r_1 + 3 = 0 \qquad r_2 + 3 = 0$$

$$r_1 = -3 \qquad r_2 = -3$$

These are real-numbered equal roots, so we call them equal real roots. Therefore, the general solution to the second order equation is

$$y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$$

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$$y(x) = c_1 e^{-3x} + c_2 x e^{-3x}$$

Finally, we'll look at one more example, this time with a characteristic equation that produces complex conjugate roots.

Example

Find the general solution to the second order equation.

$$y'' + 2y' + 17y = 0$$

The characteristic equation associated with the differential equation is

$$r^2 + 2r + 17 = 0$$

The left side doesn't easily factor, so we'll apply the quadratic formula (we could also complete the square and get the same result).

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{-2 \pm \sqrt{2^2 - 4(1)(17)}}{2(1)}$$

$$r = \frac{-2 \pm \sqrt{-64}}{2}$$

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Now we'll use the imaginary number to rewrite the root of the negative number.

$$r = \frac{-2 \pm \sqrt{(64)(-1)}}{2}$$

$$r = \frac{-2 \pm 8\sqrt{-1}}{2}$$

$$r = \frac{-2 \pm 8i}{2}$$

$$r = -1 \pm 4i$$

These are complex-numbered roots, so we call them complex conjugate roots. If we match up $r = \alpha \pm \beta i$ with $r = -1 \pm 4i$, we identify $\alpha = -1$ and $\beta = 4$, and the general solution is therefore

$$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$y(x) = e^{-x}(c_1 \cos(4x) + c_2 \sin(4x))$$

Initial value problems

With all of these examples, we've found the roots of the characteristic equation and plugged them into the general solution. What we haven't done is solve for the values of the constants c_1 and c_2 that remain in the general solution.

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The only way to find these values is with initial conditions. Because we're dealing with second order equations, we'll need two initial conditions, one for the general solution $y(x_0) = y_0$, and another for its derivative $y'(x_1) = y_1$.

Because we get an initial condition that corresponds to the derivative of the general solution, we'll need to differentiate the general solution. Once we have both the general solution and its derivatives, we'll plug in both initial conditions in order to get a system of simultaneous equations that we can solve for c_1 and c_2.

Let's try an example of an initial value problem when we have a characteristic equation with distinct real roots.

Example

Find the solution to the second order equation, if $y(0) = 5$ and $y'(0) = 27$.

$$3y'' + 18y' + 15y = 0$$

The characteristic equation associated with this differential equation is

$$3r^2 + 18r + 15 = 0$$

$$(3r + 3)(r + 5) = 0$$

The roots are therefore

$$3r_1 + 3 = 0 \qquad r_2 + 5 = 0$$

$$3r_1 = -3 \qquad r_2 = -5$$

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$$r_1 = -1$$

With these distinct real roots, the general solution is

$$y(x) = c_1 e^{-x} + c_2 e^{-5x}$$

and its derivative is

$$y'(x) = -c_1 e^{-x} - 5c_2 e^{-5x}$$

We'll substitute the initial condition $y(0) = 5$ into $y(x)$,

$$5 = c_1 e^{-(0)} + c_2 e^{-5(0)}$$

$$5 = c_1(1) + c_2(1)$$

$$5 = c_1 + c_2$$

and the condition $y'(0) = 27$ into the derivative.

$$27 = -c_1 e^{-(0)} - 5c_2 e^{-5(0)}$$

$$27 = -c_1(1) - 5c_2(1)$$

$$27 = -c_1 - 5c_2$$

Now we have a system of equations.

$$5 = c_1 + c_2$$

$$27 = -c_1 - 5c_2$$

Rewrite $5 = c_1 + c_2$ as $c_1 = 5 - c_2$, then substitute this value for c_1 into $27 = -c_1 - 5c_2$.

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$$27 = -(5 - c_2) - 5c_2$$

$$27 = -5 + c_2 - 5c_2$$

$$32 = -4c_2$$

$$c_2 = -8$$

Plugging $c_2 = -8$ into $c_1 = 5 - c_2$ gives

$$c_1 = 5 - (-8)$$

$$c_1 = 13$$

Therefore, the general solution, subject to the conditions $y(0) = 5$ and $y'(0) = 27$, is

$$y(x) = c_1 e^{-x} + c_2 e^{-5x}$$

$$y(x) = 13e^{-x} - 8e^{-5x}$$

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